

Raaeel Fatima

Backend AI Engineering Intern

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Profile Summary

Software Engineering student at MCS NUST with hands-on experience in backend development, REST APIs, databases, authentication, containerization, and introductory machine learning. Currently completing a Backend AI Engineering internship at FlyRank AI, working with Node.js, Express, SQLite, PostgreSQL, Docker, Supabase Auth, and Swagger/OpenAPI. Experienced in building CRUD APIs, implementing protected routes, working with persistent databases, and documenting APIs. Also possess experience with Python-based data analysis and linear regression projects, alongside frontend development and UI/UX design.

Education

Military College of Signals, NUST

Bachelors in Software Engineering

Rawalpindi, Pakistan

Oct. 2024 – May 2028

Punjab College, Hill House Complex

Intermediate (ICS)

Islamabad, Pakistan

2022 – 2024

Siddeeq Public School

Matriculation

Rawalpindi, Pakistan

2020 – 2022

Professional Experience

FlyRank AI

Backend AI Engineering Intern

Remote

July 2026 – Present

- Developed RESTful CRUD APIs using Node.js and Express for task management, implementing create, read, update, and delete operations.
- Implemented persistent task storage using SQLite and better-sqlite3, replacing in-memory data with a local relational database.
- Containerized backend services using Docker and Docker Compose, and configured PostgreSQL for persistent database-backed application development.
- Implemented authentication and protected API routes using Supabase Auth and JWT-based authorization, with Swagger/OpenAPI documentation for API testing and reference.

Developer's Hub

Frontend Developer Intern

Remote

2026

- Developed responsive web interfaces using HTML5, CSS3, and JavaScript, focusing on reusable layouts, structured components, and responsive design.
- Built and refined frontend modules and utility web pages while applying Git-based version control and standard web development practices.

Key Projects

CRUD REST API | *Node.js, Express, SQLite, REST API* | [\[github\]](#)

- Built a RESTful CRUD API using Node.js and Express for managing task records through structured HTTP endpoints.
- Integrated SQLite with better-sqlite3 for persistent data storage and implemented database operations for task creation, retrieval, updating, and deletion.
- Documented API endpoints using Swagger/OpenAPI and tested request-response workflows during development.

Authenticated Backend API | *Node.js, Express, Supabase Auth, JWT, Swagger* | [\[github\]](#)

- Implemented user authentication workflows including sign-up, login, and logout using Supabase Auth.
- Protected backend routes through JWT verification and authorization middleware to restrict access to authenticated users.
- Documented authentication and protected API endpoints using Swagger/OpenAPI for easier testing and integration.

Containerized PostgreSQL Backend | *Docker, Docker Compose, PostgreSQL, Node.js* | [\[github\]](#)

- Containerized backend development environments using Docker and Docker Compose with PostgreSQL as the relational database.
- Configured database services, environment variables, persistent volumes, and application-to-database connectivity for local backend development.

NUST-CASE Designer Tool | *Figma, Software Architecture, UML* | [\[casetool\]](#)

- Designed a conceptual client-server CASE tool supporting requirement workflows, UML diagramming, and automated code-generation concepts.
- Applied software architecture concepts including client-server architecture, MVC, and DAO patterns during system design.

Linear Regression & Data Mining Models | *Python, Jupyter, Data Mining* | [\[github\]](#)

- Implemented linear regression models in Python and Jupyter Notebooks to analyze relationships within structured datasets and evaluate predictive performance.
- Worked with data preprocessing and exploratory analysis workflows as part of introductory data mining and machine learning tasks.

Mini OS Kernel Simulation | *C++, System Programming, Operating Systems* | [\[github\]](#)

- Developed an object-oriented operating system kernel simulation in C++ demonstrating process scheduling and basic thread synchronization concepts.
- Implemented scheduling mechanisms including Round-Robin process handling to demonstrate core operating system concepts.

Inventory Management System | *C++, Data Design and Analysis* | [\[github\]](#)

- Developed an inventory management system in C++ for organizing, tracking, and managing product records through structured data operations.
- Applied data design and analysis concepts to manage inventory information and support efficient product record handling.

E-Commerce Store | *C++, HTML, CSS, JavaScript* | [\[github\]](#)

- Developed an e-commerce system combining a responsive frontend with C++-based backend functionality for managing product and transaction-related operations.
- Implemented structured product browsing and simulated online transaction workflows using modular application components.

Leadership, Media & Graphic Design Experience

Software Society, MCS NUST

Rawalpindi, Pakistan

Secretary Media (Incoming Term) | Deputy Director Media | Executive Member

2024 – Present

- Contributed to major university events including DEVCON 2025, DEVCON 2026, and CodeQuest through media production, event coverage, and digital communication.
- Served as Deputy Director Media for DEVCON 2026, coordinating branding strategies, digital marketing collateral, and visual asset production.
- Served as Team Media Member for DEVCON 2025, contributing to event coverage, visual communication, and creative content development.
- Nominated to serve as Secretary Media for the upcoming executive term.

SE-31 Batch Official Handle

MCS NUST

Lead Social Media Manager & Graphic Designer

2024 – Present

- Manage the official SE-31 batch social media handle, creating visual content, announcements, promotional posts, and digital media layouts.

NIC-MCS Collaboration, LNDC & MCS MUN

NUST

Team Media Lead / Creative Designer

2025 – Present

- Designed event graphics, promotional themes, and digital assets for LNDC flagship events and MCS MUN.
- Supported creative asset development and partnership-focused media campaigns in collaboration with NIC-MCS.

COMPPEC 2025

EME, NUST

Campus Ambassador

2025

- Recognized as “Most Hard-Working Ambassador” among national representatives for contributions to brand outreach, event engagement, and registration drives.

Technical & Design Skills

Backend Development: Node.js, Express.js, REST APIs, CRUD Operations, JWT Authentication, Supabase Auth, Swagger/OpenAPI

Databases: SQLite, PostgreSQL, SQL, better-sqlite3

Programming Languages: Python, JavaScript, C++, HTML5, CSS3

DevOps & Tools: Docker, Docker Compose, Git, GitHub, VS Code, Jupyter Notebooks

Machine Learning & Data: Linear Regression, Data Mining, Data Analysis

Software Engineering: SDLC, Software Architecture, UML, MVC, DAO, Operating Systems

UI/UX & Design: Figma, Canva, Wireframing, Prototyping, Typography, Event Branding